

Image tensors and channels

Project: Image-pattern Network Investigation

1. Prepare the purpose, prerequisites and materials

Teach the learning cycle

Primary target: Read a pixel using declared row column and channel axes

Session · Sequence and minutes · Resources

Session: Session 1 · Understand · 45 min

Sequence and minutes: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Sequence and minutes: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Sequence and minutes: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building. Repeated copies of a case share one recorded attempt; they are not new evidence.

Retrieval answer guidance

Learned representations

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an input through a hidden representation and a decision rule

Reference example, not the only acceptable wording: Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. Predictions are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

Initialisation and reproducibility

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

Look for: Trace how starting values affect a fixed training budget

Reference example, not the only acceptable wording: Teacher explanation: States are -2, 0 and 1. Final training MSE is 2.5. Each step halves the signed distance from 2.

Learning rate experiments

Without opening the old lesson, explain one key idea from Learning rate experiments. Give an example and a mistake that the idea helps you avoid.

Look for: Run a controlled comparison of learning rates

Reference example, not the only acceptable wording: Teacher explanation: First $v = 0.5$ with MSE 5.625. Second gradient -7.5 gives $v = 0.875$ and MSE 3.1640625. Distance is multiplied by 0.75 each step.

Use two 45-minute sessions. Draw the four-pixel image and label row, column and channel before writing nested brackets. Keep RGB ordering, zero-based indices and the 0 to 255 contract explicit. Distinguish an axis permutation from regrouping, and fixed scaling from fitted preprocessing. No external image files or student photographs are needed.

Outcomes

Read a pixel using declared row column and channel axes

Convert channel layout without changing pixel meaning

Distinguish fixed value scaling from information losing transformations

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Different inputs can map to the same representation. The number of components alone does not guarantee that the original values can be recovered.

End goal for the pictorial case

By the end, I can explain image tensors and channels using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

2. Explain and model the mechanism

A small fictional RGB image has height 2 pixels, width 2 pixels and three channels ordered red, green, blue. Store it as `image[row][column][channel]`, abbreviated HWC. Use zero-based indices: the top row is 0, the left column is 0, and red, green and blue are channels 0, 1 and 2. Each stored component is an integer from 0 to 255.

The shape is 2 by 2 by 3. There are four pixels and twelve channel values. The blue component of the bottom-left pixel is `image[1][0][2] = 255`. A single channel is a plane of values across positions; it is not a separate photograph or a label.

Shape needs a schema. A grayscale image might be stored as H by W or H by W by 1. An RGB image uses three channels here; other formats can include additional channels. Declare the format instead of assuming every three-dimensional array means the same thing.

A channel-first layout uses `image[channel][row][column]`, abbreviated CHW. Our image then has shape 3 by 2 by 2. A correct conversion preserves the rule `chw[c][r][k] = hwc[r][k][c]`. The code assumes a nonempty rectangular image with exactly three values per pixel.

Reading nested values in order and merely regrouping them into a new shape does not perform this axis conversion. The first four flattened HWC values are `[255, 0, 0, 0]`. A naive regrouping makes these the first channel plane, but the correct red plane is `[[255, 0], [0, 255]]`. Shape agreement alone cannot detect this mistake.

Term · Meaning

Term: Tensor

Meaning: A numerical array with a declared set of axes

Term: Channel

Meaning: One component plane of an image such as red green or blue

Term: Axis order

Meaning: The sequence of meanings assigned to array dimensions

Term: Value scaling

Meaning: A declared numerical conversion applied to stored values

The same pixels in three channel planes

Channel · Top row · Bottom row

Channel: Red c 0

Top row: [255, 0]

Bottom row: [0, 255]

Channel: Green c 1

Top row: [0, 255]

Bottom row: [0, 255]

Channel: Blue c 2

Top row: [0, 0]

Bottom row: [255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Guided: top-right is [0, 255, 0] and bottom-right is [255, 255, 255]. The top-right green value is `hwc[0][1][1]` or `chw[1][0][1]`, both 255. Exit: shape alone misses both errors. Check the top-left RGB pixel is red before scaling and that a raw value 255 becomes exactly 1 under one declared division, rather than approximately 0.00392.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a prediction, then explain the deciding step. These teacher notes contain solutions; do not reveal the independent case answer during modelling.

The same pixels in three channel planes

Channel · Top row · Bottom row

Channel: Red c 0

Top row: [255, 0]

Bottom row: [0, 255]

Channel: Green c 1

Top row: [0, 255]

Bottom row: [0, 255]

Channel: Blue c 2

Top row: [0, 0]

Bottom row: [255, 255]

Each row describes one channel plane. Within a plane the first listed row is the image top and the second is the bottom. Combining the three values at the same position recovers its RGB triplet.

Use the planes to reconstruct the top-right and bottom-right RGB pixels. Give both HWC and CHW indices for the top-right green value.

Thinking move · Teacher prompt

Thinking move: Plan

Teacher prompt: What is given? What is the question? Which procedure fits these facts?

Thinking move: Do

Teacher prompt: Point to each input and intermediate step in the worked trace. Name the rule that allows

the next step.

Thinking move: Check

Teacher prompt: Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Image tensors and channels

Move colour values between tensor layouts without changing the pixels.

The facts we will use

Toy RGB image has 2 rows, 1 column. Top pixel [1, 4, 7]; bottom [2, 5, 8]. HWC means height, width, channel.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/20

Two RGB pixels before layout conversion

	red	green	blue
top pixel	1	4	7
bottom pixel	2	5	8

Read a cell using both its row and column labels.

Image tensors and channels

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Record HWC shape

Shape is [2, 1, 3]; top green is 4 and bottom green 5.



2. Reorder to CHW

Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].



3. Check same pixel

Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

CHANGE ONE THING

Add a batch containing three identical images in NHWC layout.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Record HWC shape

What happens and why: Shape is [2, 1, 3]; top green is 4 and bottom green 5.

Step: 2. Reorder to CHW

What happens and why: Shape [3, 2, 1]; red plane [[1],[2]], green [[4],[5]], blue [[7],[8]].

Step: 3. Check same pixel

What happens and why: Bottom green is HWC [1, 0, 1]=5 and CHW [1, 1, 0]=5.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

A tensor is an array with named dimensions. HWC orders height, width and colour; CHW puts colour first. Rearranging the layout changes where values are stored, not the colour measurement itself. Trace a named pixel through both layouts using zero-based indices. Adding a batch dimension counts images and does not add a new colour channel.

What this example does not establish

Shape consistency does not prove correct colour order or value range; document channel names and preprocessing.

Change one thing in the pictured case

Change: Add a batch containing three identical images in NHWC layout.

Prediction: Batch shape is [3, 2, 1, 3].

Reason: The new first dimension counts images; it does not add a fourth colour channel.

Ask while showing the picture

Cover the result. Ask learners to name the inputs and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its evidence. Then change the supplied condition and compare. The picture is a representation; it does not document execution of a real AI model.

3. Demonstrate and guide practice

Time or stage · Teacher action

Time or stage: Session 1 start to 7 minutes

Teacher action: Retrieve information loss and label the four image positions with zero-based indices.

Time or stage: 7 to 18 minutes

Teacher action: Count pixels and components, then use W1 to read a full pixel and one channel.

Time or stage: 18 to 32 minutes

Teacher action: Trace the three channel planes and run or hand-trace the optional code.

Time or stage: 32 to 45 minutes

Teacher action: Use W2 to expose naive regrouping with the bottom-right red component.

Time or stage: Session 2 start to 8 minutes

Teacher action: Reconstruct the guided pixels and locate the same green value in both layouts.

Time or stage: 8 to 20 minutes

Teacher action: Use W3 to compare fixed scaling with colour averaging and discuss double scaling.

Time or stage: 20 to 35 minutes

Teacher action: Collect W4 independently, checking the non-square shape and the leading batch axis.

Time or stage: 35 to 45 minutes

Teacher action: Use W5 and the exit response to plan exact-value checks, then administer the fresh retry.

Use the worked example in Sections 3 and 6. Collect a prediction before revealing its result; ask the learner to point to each intermediate fact.

Use three labelled paper grids and a coordinate ledger; provide colour names so success does not depend on colour vision. Allow fraction answers for scaling. Extension: convert a rectangular image HWC to CHW and back, checking every coordinate, then construct two different colours with the same classroom channel mean.

Reduce help deliberately

Completion task: A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Teacher check: Teacher explanation: The incorrect plane is $\begin{bmatrix} 255 & 0 \\ 0 & 0 \end{bmatrix}$. The correct red plane is $\begin{bmatrix} 255 & 0 \\ 0 & 255 \end{bmatrix}$. They disagree at row 1 column 1. Regrouping consecutive values does not transpose axes.

Start by identifying the given inputs together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood. For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Image tensors and channels. Proposed explanation: “Three channels mean three images”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Channels are component planes at matching spatial positions in one image.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class. If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Where is the bottom green pixel in the first batched image? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Reasoned teacher answer

NHWC [0, 1, 0, 1]=5: image0, row1, column0, green channel1. The value is unchanged by batching.

Changed-case check: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour channel.

Changed-case reasoning: Batch shape is [3, 2, 1, 3]. The new first dimension counts images; it does not add a fourth colour channel.

Collect each learner’s explanation before discussion. Use the first missing connection between input, deciding step and result to choose support. This question is worked-case practice, not fresh mastery evidence.

4. Run the investigation and diagnose misconceptions

Three channels mean three images: Channels are component planes at matching spatial positions in one image.

Reshaping automatically changes axis meanings correctly: Axis permutation must preserve coordinate correspondence; regrouping a flattened list can mix positions and colours.

Every conversion to a smaller number preserves information: Exact fixed scaling differs from rounding and channel averaging, which can collapse distinct values.

Use Section 9 for the evidence ledger. Add a boundary or missing-input case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows · Next teaching action

What the response shows: Missing or misread facts

Next teaching action: Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows: Wrong procedure

Next teaching action: Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows: Arithmetic or implementation error

Next teaching action: Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows: Correct result without a reason

Next teaching action: Ask for a trace and a changed case before marking independent understanding.

What the response shows: An unsupported wider claim

Next teaching action: Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal. Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

5. Assess and explain the answers

Guided and exit questions

Guided: top-right is [0, 255, 0] and bottom-right is [255, 255, 255]. The top-right green value is `hwc[0][1][1]` or `chw[1][0][1]`, both 255. Exit: shape alone misses both errors. Check the top-left RGB pixel is red before scaling and that a raw value 255 becomes exactly 1 under one declared division, rather than approximately 0.00392.

W1 Read a pixel precisely

For the lesson image, give `image[1][0]`, `image[1][0][2]`, the HWC shape, pixel count and number of channel values.

Teacher explanation: The RGB pixel is [0, 0, 255]; its blue value is 255. Shape is 2 by 2 by 3, with four pixels and twelve channel values.

W2 Audit an axis conversion

A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Teacher explanation: The incorrect plane is $[[255, 0], [0, 0]]$. The correct red plane is $[[255, 0], [0, 255]]$. They disagree at row 1 column 1. Regrouping consecutive values does not transpose axes.

W3 Check scaling and lost colour

Scale $[255, 128, 0]$ by 255, keeping the middle value as a fraction. Then calculate the classroom RGB means of pure red and pure green. Which operation loses the distinction between those colours?

Teacher explanation: Scaling gives $[1, 128/255, 0]$. Both RGB means are 85, so averaging loses the red versus green distinction. Exact fixed scaling preserves the triplet distinction.

W4 Convert a new image independently

A one-row two-column RGB image has pixels $[10, 20, 30]$ then $[40, 50, 60]$. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts. A batch contains four copies: give its NHWC shape and total channel-value count.

Teacher explanation: HWC shape is 1 by 2 by 3. CHW planes are $[[10, 40]]$, $[[20, 50]]$, $[[30, 60]]$, giving shape 3 by 1 by 2. $\text{hwc}[0][1][1]$ and $\text{chw}[1][0][1]$ both equal 50. Batch shape is 4 by 1 by 2 by 3, containing 24 values.

W5 Design revealing pipeline checks

The lesson image is loaded with red and blue exchanged, then scaled twice. Give two expected checks that would expose these errors even though the shape is correct.

Teacher explanation: Before scaling, the top-left pixel must be $[255, 0, 0]$, not $[0, 0, 255]$. A known raw component of 255 must become 1 after scaling once; approximately 0.00392 indicates division twice. Test named values and positions as well as dimensions.

Independent assessment and fresh retry

W4 requires three planes, both coordinate references to value 50, NHWC shape 4 by 1 by 2 by 3 and 24 values. Fresh retry: a two-row one-column image contains $[3, 6, 9]$ above $[12, 15, 18]$. HWC shape is 2 by 1 by 3. CHW planes are $[[3], [12]]$, $[[6], [15]]$, $[[9], [18]]$. The bottom blue component is $\text{hwc}[1][0][2] = \text{chw}[2][1][0] = 18$. A batch of three has NHWC shape 3 by 2 by 1 by 3 and 18 values.

Assess independent transfer

Selected case: A one-row two-column RGB image has pixels $[10, 20, 30]$ then $[40, 50, 60]$. State its HWC shape and all three CHW planes. Give the green value at row 0 column 1 in both layouts. A batch contains four copies: give its NHWC shape and total channel-value count.

Selected case answer

Teacher explanation: HWC shape is 1 by 2 by 3. CHW planes are `[[10, 40]]`, `[[20, 50]]`, `[[30, 60]]`, giving shape 3 by 1 by 2. `hwc[0][1][1]` and `chw[1][0][1]` both equal 50. Batch shape is 4 by 1 by 2 by 3, containing 24 values.

Dimension · Evidence required

Dimension: Explanation

Evidence required: Names the applicable idea or procedure and ties it to the supplied facts.

Dimension: Trace and result

Evidence required: Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension: Limits

Evidence required: States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace. Do not penalise an honestly recorded model failure if the learner correctly explains its evidence and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case. Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

6. Connect the project, support and next lesson

Add a labelled image tensors and channels evidence sheet to Image-pattern Network Investigation. Show how you can read a pixel using declared row column and channel axes. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use three labelled paper grids and a coordinate ledger; provide colour names so success does not depend on colour vision. Allow fraction answers for scaling. Extension: convert a rectangular image HWC to CHW and back, checking every coordinate, then construct two different colours with the same classroom channel mean.

Week 21 introduces convolution as a local operation with explicit neighbourhoods and boundary rules.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/21, ai-5/23, ai-5/26. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used. For a new level, revisit the previous capstone before checking the new prerequisites.

Record · Teacher evidence

Record: Learner code and date

Teacher evidence: _____

Record: Concept and case used; prior exposure

Teacher evidence: _____

Record: Explanation / trace / limits

Teacher evidence: _____

Record: Support and first repair

Teacher evidence: _____

Record: Changed case and delayed outcome

Teacher evidence: _____

Record: Next teaching action

Teacher evidence: _____

Use the evidence to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

7. Prepare, teach and assess

Teaching target: Read a pixel using declared row column and channel axes

Prepare the labelled worked-case picture, the matching workbook, fictional input cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

Session: Session 1 · Understand · 45 min

Plan: 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

Resources: Sections 1–6; record the brief recall in the workbook.

Session: Session 2 · Practise · 45 min

Plan: 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

Resources: Sections 7–10; use the matching workbook W2 and misconception check.

Session: Session 3 · Investigate and transfer · 45 min

Plan: 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.

Resources: Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: A program flattens the lesson HWC array and groups the first four values into a 2 by 2 red plane. Give that incorrect plane and the correct plane. Identify a position where they disagree.

Reasoned practice answer: Teacher explanation: The incorrect plane is $[[255, 0], [0, 0]]$. The correct red plane is $[[255, 0], [0, 255]]$. They disagree at row 1 column 1. Regrouping consecutive values does not transpose axes.

Check every learner before discussion: Use the worked case for Image tensors and channels. Proposed explanation: “Three channels mean three images”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Channels are component planes at matching spatial positions in one image.

Support: read the exact case aloud, enlarge the labelled picture, supply input cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Add a batch containing three identical images in NHWC layout. Require a prediction and a reason before comparing. Expected result: Batch shape is [3, 2, 1, 3]. Reason: The new first dimension counts images; it does not add a fourth colour channel.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion · Not yet · With support · Independent

Criterion: Explanation

Not yet: Names an answer without a deciding fact

With support: Uses a hint to link fact and decision

Independent: Explains the deciding fact in own words

Criterion: Trace or test

Not yet: Cannot connect input and result

With support: Completes steps with prompts

Independent: Traces or tests a changed case accurately

Criterion: Limits and revision

Not yet: Claims more than evidence supports

With support: Names a limit after a prompt

Independent: States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner’s explanation and evidence.

8. Use feedback, labs and repair paths

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding evidence rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual outputs; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as evidence. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.